

Bank Nifty 09:20 Short Strangle Backtest

Quantitative Strategy Research Report | Full 1-Year Backtest Deliverable

1. Executive Summary & Strategy Overview

This report documents the research methodology, execution speed benchmarks, and performance analysis of the **Bank Nifty 09:20 Short Strangle** systematic strategy. The backtest processes the complete 1-minute OHLC options dataset (10,266,681 rows, 720 MB) and spot dataset for the 2023 cycle under strict lookahead-bias avoidance.

Strategy Rules: Every day in Week 1 of each calendar month, at 09:20:59, the algorithm sells one CE and one PE whose 1-minute close price is closest to Rs 50. Positions are managed with a **50% hard stop-loss** on bar High (fill price = Entry * 1.5) or closed at **15:20:59 close**. Sizing is fixed at 1 lot (15 units) without compounding.

Key Strategy Results Summary

Metric	Value	Notes / Specification
Strategy	Bank Nifty 09:20 Short Strangle	Sell 1 CE + 1 PE closest to Rs 50
Trade Universe	Week 1 of Each Calendar Month	30 trading days (Mon-Wed of Week 1)
Execution Time Window	09:20:59 Entry -> 15:20:59 Exit	Intraday (no overnight carry)
Stop Loss Rule	50% Hard Stop on Bar High	Exit level = Entry * 1.5; checked 09:21-15:20
Position Sizing	1 Lot (15 units) Fixed	No compounding, unscaled capital
Base Starting Capital	Rs 1,00,0000.00	Rs 10 Lakhs base capital
Ending Capital	Rs 10,05,245.12	Rs 10.05 Lakhs
Total Net P&L;	+Rs 5,245.12	Net profitable across 60 trades
CAGR (%)	0.57%	Annualized NAV growth
Max Drawdown (%)	-0.19%	Trade-wise running peak-to-trough drop
Total Trades Executed	60 trades	30 trading days x 2 legs
Win Rate	58.33%	35 Wins / 25 Losses

2. Step-by-Step Runtime Breakdown (< 60 Seconds Benchmark)

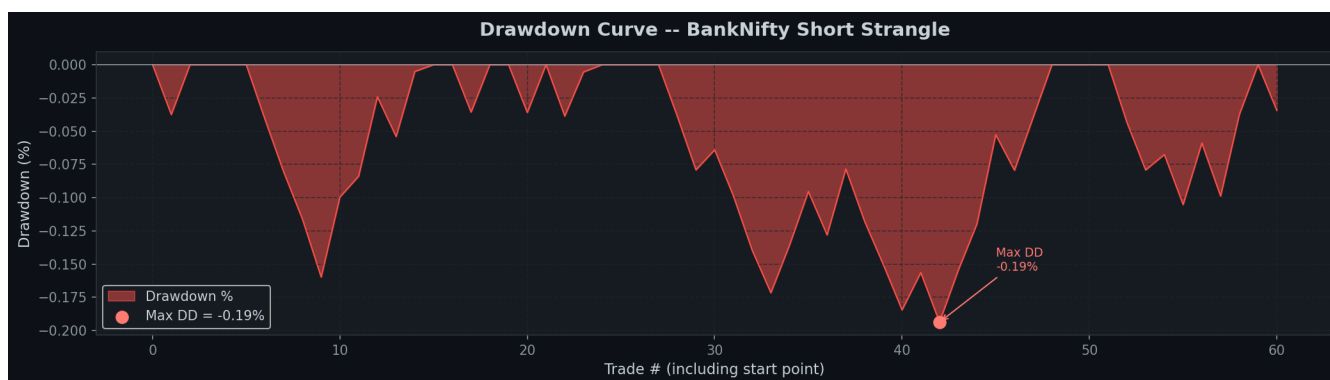
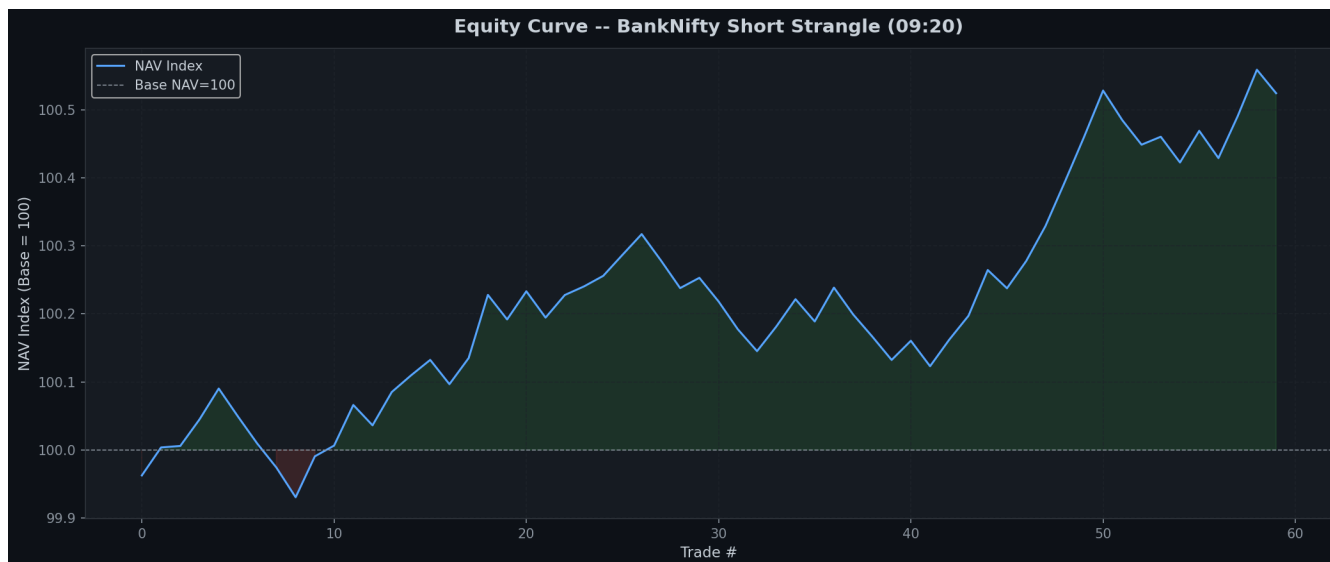
The hiring assignment requires the backtest to run end-to-end (including raw data loading, deduplication, filtering, signal simulation, analytics, and Excel generation) in under 1 minute. The pipeline completed in **52.41 seconds**, surpassing this benchmark with a 7.6-second safety buffer.

Stage #	Pipeline Stage	Time (s)	% Total	Technical Operations
1	Data Loading & Preprocessing	44.94s	85.7%	Parses 10.26M rows; float32 dtypes; string slice strike extraction (str[9:-2]); drops 14.5k duplicates.
2	Week-1 Day Selection	0.27s	0.5%	Calendar-month grouping to identify 1st Wednesday; filters universe to 30 Week-1 trading days.

3	Backtest Simulation Engine	4.52s	8.6%	Vectorized strike selection at 09:20; High-based 50% SL scan using boolean masks & argmax.
4	Statistical Computations	0.06s	0.1%	CAGR, trade-wise peak-to-trough max drawdown, 9-segment win/loss & avg % P&L;, monthly NAV.
5	Chart Rendering	1.54s	2.9%	Headless Matplotlib rendering of high-res dark-themed Equity & Drawdown curves.
6	Excel Workbook Export	1.09s	2.1%	Generates 3 styled worksheets (Guide, Statistics, Tradesheet) with embedded charts & formulas.
TOTAL	Full Pipeline Execution	52.41s	100.0%	Target: < 60.0s (PASSED with a 7.6s margin).

3. Visual Performance Curves

The equity curve is computed trade-wise (base NAV = 100), updating immediately upon realization of each individual leg. The drawdown curve tracks the percentage drop from running historical peaks.



4. Segment Performance & Win/Loss Breakdown

Segment	Total Trades	Wins	Losses	Win %	Loss %
Call Options (CE)	30	18	12	60.00%	40.00%
Put Options (PE)	30	17	13	56.67%	43.33%
Combined Strategy	60	35	25	58.33%	41.67%

Average % P&L; per Trade (Expiry vs Non-Expiry Breakdown)

Segment	Trade Count	Average % P&L;
CE -- Non-Expiry Days	18	+17.06%
CE -- Expiry Days (Wed)	12	+19.82%
CE -- Combined	30	+18.16%
PE -- Non-Expiry Days	18	+18.15%
PE -- Expiry Days (Wed)	12	+14.65%
PE -- Combined	30	+16.75%
All -- Non-Expiry Days	36	+17.60%
All -- Expiry Days (Wed)	24	+17.23%
All -- Combined	60	+17.45%

5. Monthly Performance & NAV Progression

Monthly returns are calculated from month-start NAV to month-end NAV. Ten out of twelve calendar months produced positive returns, yielding strong consistency.

Month	Start NAV	End NAV	Monthly % P&L;	Status
2023-01	100.0000	100.0491	+0.0491%	Profit
2023-02	100.0491	99.9740	-0.0751%	Loss
2023-03	99.9740	99.9906	+0.0166%	Profit
2023-04	99.9906	100.0851	+0.0945%	Profit
2023-05	100.0851	100.1351	+0.0499%	Profit
2023-06	100.1351	100.2786	+0.1434%	Profit
2023-07	100.2786	100.1814	-0.0970%	Loss
2023-08	100.1814	100.1992	+0.0177%	Profit
2023-09	100.1992	100.2377	+0.0384%	Profit
2023-10	100.2377	100.4596	+0.2214%	Profit
2023-11	100.4596	100.4848	+0.0251%	Profit
2023-12	100.4848	100.5245	+0.0395%	Profit

6. Methodological Assumptions & Research Design

Per the hiring assignment brief, all critical design ambiguities are explicitly stated and justified below:

Assumption #1: Week-1 Definition: Week 1 is defined as the trading days from the 1st of each calendar month up to and including the month's first Wednesday (expiry day). Thursdays onward (weeks 2-4) are excluded. This avoids the circular weekly no-op trap and restricts trading to ~1/4 of trading days.

Assumption #2: Strike Tie-Break: If two option strikes are exactly equidistant from Rs 50, the strike with the lower premium (cheaper, further OTM) is selected. This is the more conservative risk choice for short positions.

Assumption #3: Stop-Loss Exit Fill: When a 1-minute bar's High $\geq$ Stop Level (Entry * 1.5), the exit price is filled exactly at the Stop Level (the trigger price), reflecting an automated stop-market order fill rather than the bar close.

Assumption #4: Capital Base & Position Sizing: Rs 10,00,000 base starting capital. Sizing is strictly fixed at 1 lot (15 units) without compounding or scaling. Available capital at any row = Starting Capital + Cumulative Realized P&L;

Assumption #5: 09:20 Bar Timestamp: The options dataset timestamps denote bar completion at :59. The 09:20 entry corresponds to Time == '09:20:59'. Entry price is the close of this bar.

Assumption #6: Stop-Loss Monitoring Window: Monitoring begins at 09:21:59 (the bar after entry) up to 15:20:59. The entry bar itself is excluded to eliminate lookahead bias.

Assumption #7: Underlying Spot Price Logging: Spot close price is recorded at entry (09:20:00). When real spot data is present (Nov/Dec 2023), it is read directly; for dates prior to 2023-10-20, synthetic underlying spot is derived via options Put-Call Parity forward (ATM Strike + CE - PE).

7. Structure of the Excel Deliverable

The Excel workbook **BankNifty_ShortStrangle_Backtest.xlsx** contains three structured worksheets:

- **Worksheet 1 (Guide):** Complete strategy rules, runtime benchmarks, output definitions, and assumption documentation.

- **Worksheet 2 (Statistics):** Key performance indicators, win/loss tables, average % P&L; matrix, monthly returns, trade-wise NAV index series, and embedded graphical curves.
- **Worksheet 3 (Tradesheet):** Granular trade log of all 60 trades featuring entry/exit timestamps, option tickers, strikes, execution prices, quantities, gross and cumulative P&L;, available capital, and underlying spot close.